

NATURAL FREQUENCIES OF FRAMES WITH AXIALLY LOADED TIMOSHENKO MEMBERS

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(Received 12 June 1972, and in revised form 16 November 1972)

A convenient dynamic stiffness matrix method for converging with certainty upon any particular natural frequency of a plane frame is presented. The constituent members are considered to have a uniform distribution of mass and the effects of axial load, rotatory inertia and shear deflection, uniquely described by the parameters p , r and s , respectively, may be accounted for. Any particular effect can be neglected by setting the relevant parameter equal to zero.

Experimental and theoretical results for the first four in-plane, flexural natural frequencies of an H-shaped frame, for various column lengths and axial loads, are presented. Theory and experiment agree fairly well.

1. INTRODUCTION

The formation of dynamic member stiffness equations, relating the sinusoidally varying end forces and displacements for each constituent member of a structure, provides the basis of a convenient method for converging upon the natural frequencies of any plane frame. From these member equations the overall dynamic stiffness matrix of the frame, $\mathbf{K}$, can be assembled. The natural frequencies then correspond to the values of ω , the circular frequency, which satisfy the matrix equation

$$\mathbf{K}\mathbf{D} = \mathbf{0}, \quad (1)$$

where $\mathbf{D}$ is the vector of the amplitudes of the sinusoidally varying nodal displacements, and $\mathbf{K}$ is a function of ω .

This paper presents such an approach to determining the undamped natural frequencies of any plane frame, in which also a distributed mass technique is used and the effects are incorporated of axial load, rotatory inertia and shear deflection, which are ignored in the classical Bernoulli–Euler theory of flexural vibrations. Results obtained for a simple structure are compared with experimental results in section 3.

Recently several authors have used Bernoulli–Euler vibration theory, assuming either a uniform distribution of mass [1, 2] or concentrated masses [3], in conjunction with equation (1). The more sophisticated Timoshenko theory, which includes the effects of rotatory inertia and shear deflection, has been used extensively. For instance, Traill-Nash and Collar [4] and Huang [5] have given frequency equations for single beams with various boundary conditions while Henshell and Warburton [6], Cheng [7] and Wang and Kinsman [8] have tackled frameworks. The effect of axial load alone has also been assessed for beams and frameworks for which the effects of shear deflection and rotatory inertia are ignored [9]. Finally, Szidarovszky [10], and later Djodjo [11], have allowed for the combined effects of axial load, rotatory inertia and shear deflection while, respectively, considering a simply supported beam and a transfer matrix approach to a beam on elastic foundations.

Converging on the natural frequencies of a structure considered to have a uniform distribution of mass presents a number of problems. Unlike a lumped mass system, where the number of natural frequencies is finite and known, an infinite number of natural frequencies exist. Several of these natural frequencies may be close together or coincident, while others may correspond to $D = 0$. Also as $|K|$ is now a highly irregular function of ω any trial and error method, which involves computing $|K|$ and observing when it changes sign, may miss roots. These problems can be completely overcome by the use of an algorithm based on a theorem due to Lord Rayleigh [12]. This algorithm [13] states that

$$J = J_0 + s\{K\}, \quad (2)$$

where J represents the number of natural frequencies of the structure exceeded by some trial frequency, ω^* , J_0 represents the number of natural frequencies which would still be exceeded if constraints were imposed upon the structure so as to make all the nodal displacements D zero, and $s\{K\}$ is the sign count of the matrix K . $s\{K\}$ is defined in reference [13], and is equal to the number of negative elements on the leading diagonal of the upper (or lower) triangular matrix obtained from K , when $\omega = \omega^*$, by the Gaussian elimination procedure in the following form. The rows of K are taken as pivotal in order, and suitable multiples of each pivotal row are added to all following rows, so that all elements in the pivotal column, which are below the pivot, become zero.

Clearly $s\{K\}$ is easily computed. The computation of J_0 is dealt with at the end of section 2.

2. MEMBER THEORY

In-plane vibration of a uniform beam-column is now considered, with axial load, rotatory inertia and shear deflection included. The dynamic member stiffness matrices, which are needed when assembling K , are thus obtained. The procedure for computing J_0 is then considered in depth.

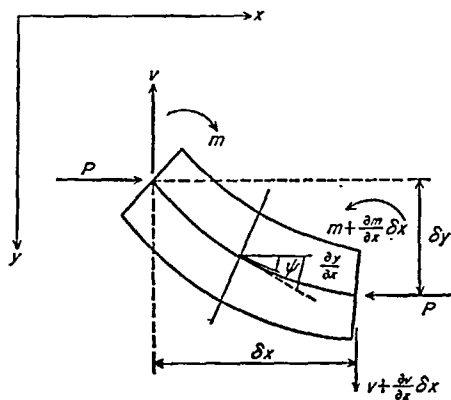


Figure 1. Forces acting on a typical element.

Figure 1 shows a typical element of any uniform beam-column undergoing flexural vibrations. The axial compressive load is P . (P can be positive or negative. Hence tension is included.) The positive directions of the forces and displacements are as shown. v = shear force due to bending, m = bending moment, L = length of member, A = cross-sectional area, I = second moment of area for in-plane bending, E , G and ρ are Young's modulus,

the modulus of rigidity and the density of the member material, respectively, $k' =$ section shape factor, $\psi =$ angle of rotation of the cross-section and $\partial y/\partial x =$ total slope, which is equal to the angle between the x -axis and the tangent at the neutral axis (which is not equal to ψ because of the component of deflection due to shear).[†] Hence, resolving vertically and taking moments gives

$$\frac{\partial v}{\partial x} = \rho A \frac{\partial^2 y}{\partial t^2}, \quad (3)$$

$$v = \frac{\partial m}{\partial x} - P \frac{\partial y}{\partial x} + \rho I \frac{\partial^2 \psi}{\partial t^2}, \quad (4)$$

where $\rho I \partial^2 \psi / \partial t^2$ is the rotatory inertia term. Simple bending theory gives

$$m = -EI \frac{\partial \psi}{\partial x}. \quad (5)$$

Finally consideration of the shearing of the element gives

$$\psi = \left(1 - \frac{P}{\phi}\right) \frac{\partial y}{\partial x} - \frac{v}{\phi}, \quad (6)$$

where

$$\phi = k' AG. \quad (7)$$

Equations (3)–(7) define the motion of an axially loaded Timoshenko beam. Eliminating v , m , and either y or ψ from these equations yields

$$\begin{aligned} EI \left(1 - \frac{P}{\phi}\right) \frac{\partial^4 \theta}{\partial x^4} + \rho A \frac{\partial^2 \theta}{\partial t^2} + P \frac{\partial^2 \theta}{\partial x^2} - \left[I \rho \left(1 - \frac{P}{\phi}\right) + \frac{EI \rho A}{\phi} \right] \frac{\partial^4 \theta}{\partial x^2 \partial t^2} \\ + \rho I \frac{\rho A}{\phi} \frac{\partial^4 \theta}{\partial t^4} = 0, \end{aligned} \quad (8)$$

where

$$\theta = y \quad \text{or} \quad \psi. \quad (9)$$

If a sinusoidal variation of y and ψ , with circular frequency ω , is assumed, then

$$\theta(x, t) = \Theta(x) \sin \omega t, \quad (10)$$

where

$$\Theta(x) = Y(x) \quad \text{or} \quad \Psi(x) \quad (11)$$

and $Y(x)$ and $\Psi(x)$ are the amplitudes of the sinusoidally varying displacement and bending slope, respectively.

Also let

$$\xi = x/L. \quad (12)$$

Then substituting equations (3), (10), (11) and (12) in equation (6), differentiated with respect to x , gives

$$D^{II}Y + \frac{b^2 s^2}{(1 - s^2 p^2)} Y - \frac{L}{(1 - s^2 p^2)} D\Psi = 0 \quad (13)$$

[†] A list of notation is given in Appendix II.

and substituting equations (10) and (12) in equation (8) gives

$$\left[D^{IV} + \frac{b^2 \Delta}{(1-s^2 p^2)} D^{II} - \frac{b^2(1-b^2 r^2 s^2)}{(1-s^2 p^2)} \right] \Theta = 0, \quad (14)$$

where D is the operator $d/d\xi$ and

$$\begin{aligned} b^2 &= \rho A L^4 \omega^2 / EI, \\ r^2 &= I / A L^2, \\ s^2 &= EI / \phi L^2, \\ p^2 &= P L^2 / EI, \\ \Delta &= (p^2 / b^2) + r^2(1-s^2 p^2) + s^2. \end{aligned} \quad (15)$$

General solutions for Y and Ψ may be found from equation (14). It is assumed here that $b^2 r^2 s^2 < 1$. ($b^2 r^2 s^2 > 1$ is treated in Appendix I.) Thus

$$Y = C_1 \cosh \alpha \xi + C_2 \sinh \alpha \xi + C_3 \cos \beta \xi + C_4 \sin \beta \xi, \quad (16)$$

$$\Psi = C_1' \sinh \alpha \xi + C_2' \cosh \alpha \xi + C_3' \sin \beta \xi + C_4' \cos \beta \xi, \quad (17)$$

where

$$\frac{\alpha}{\beta} = \frac{b}{[2(1-s^2 p^2)]^{\frac{1}{2}}} \left\{ \mp \Delta + \left[\Delta^2 + \frac{4}{b^2} (1-s^2 p^2) (1-b^2 r^2 s^2) \right]^{\frac{1}{2}} \right\}^{\frac{1}{2}}. \quad (18)$$

Substituting equations (16) and (17) into equation (13) gives

$$H = \frac{(1-s^2 p^2) \alpha^2 + b^2 s^2}{\alpha L}, \quad (19)$$

$$Z = \frac{(1-s^2 p^2) \beta^2 - b^2 s^2}{\beta L}, \quad (20)$$

$$\begin{aligned} C_1' &= H C_1, \\ C_2' &= H C_2, \\ C_3' &= -Z C_3, \\ C_4' &= Z C_4. \end{aligned} \quad (21)$$

The constants C_1 – C_4 and C_1' – C_4' may be found from equations (16), (17) and (21), with the following boundary conditions:

$$\begin{aligned} \text{at } \xi = 0, \quad Y &= \delta_{y1} \quad \text{and} \quad \Psi = \Psi_1; \\ \text{at } \xi = 1, \quad Y &= \delta_{y2} \quad \text{and} \quad \Psi = \Psi_2. \end{aligned} \quad (22)$$

These values of C_1 – C_4 and C_1' – C_4' can be substituted in equations (16) and (17). The required dynamic member equations, which relate the amplitudes of the sinusoidally varying end forces m_1 , m_2 , p_{y1} , p_{y2} and displacements Ψ_1 , Ψ_2 , δ_{y1} , δ_{y2} , as shown in Figure 2, may then be obtained by substituting equations (16), (17) and (21) into equations (5) and (6).

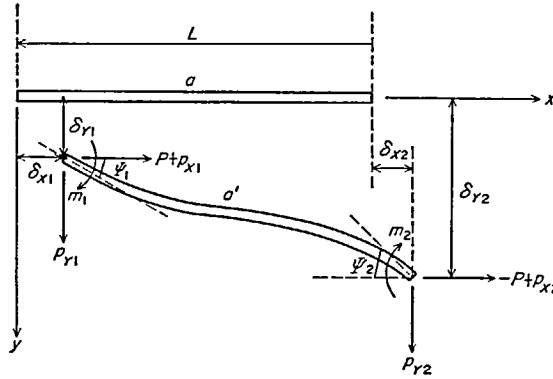


Figure 2. Forces and displacements at the ends of a member. a and a' are the undeflected and deflected positions of the member, respectively, and Ψ_1 and Ψ_2 are the rotations of the end cross-sections.

The member equations relating the longitudinal forces and displacements may be obtained from a number of standard works, such as reference [14]. They are

$$\begin{aligned} p_{x1} &= hv \cot v \delta_{x1} - hv \operatorname{cosec} v \delta_{x2}, \\ p_{x2} &= -hv \operatorname{cosec} v \delta_{x1} + hv \cot v \delta_{x2}, \end{aligned} \quad (23)$$

where

$$h = EA/L, \quad (24)$$

$$v = \omega L \sqrt{(\rho/E)}, \quad (25)$$

p_{x1} and p_{x2} are the amplitudes of the sinusoidal variations of the longitudinal forces about the mean value P and δ_{x1} and δ_{x2} are the amplitudes of the corresponding sinusoidally varying longitudinal displacements, as shown in Figure 2.

Hence, in the usual matrix notation, the complete, in-plane, dynamic member stiffness equations (referred to member axes) are as follows:

$$\begin{aligned} \mathbf{p}_1 &= \mathbf{k}_{11} \mathbf{d}_1 + \mathbf{k}_{12} \mathbf{d}_2, \\ \mathbf{p}_2 &= \mathbf{k}_{21} \mathbf{d}_1 + \mathbf{k}_{22} \mathbf{d}_2, \end{aligned} \quad (26)$$

where

$$\begin{aligned} \mathbf{d}_1 &= \{\delta_{x1}, \delta_{y1}, \Psi_1\}, & \mathbf{d}_2 &= \{\delta_{x2}, \delta_{y2}, \Psi_2\}, \\ \mathbf{p}_1 &= \{p_{x1}, p_{y1}, m_1\}, & \mathbf{p}_2 &= \{p_{x2}, p_{y2}, m_2\}, \end{aligned} \quad (27)$$

and

$$\begin{aligned} \mathbf{k}_{11} &= \begin{bmatrix} hv \cot v & 0 & 0 \\ 0 & \frac{Tk}{L^2} & \frac{Qk}{L} \\ 0 & \frac{Qk}{L} & Sk \end{bmatrix}, \\ \mathbf{k}_{22} &= \begin{bmatrix} hv \cot v & 0 & 0 \\ 0 & \frac{Tk}{L^2} & -\frac{Qk}{L} \\ 0 & -\frac{Qk}{L} & Sk \end{bmatrix}, \end{aligned}$$

$$\mathbf{k}_{12} = \mathbf{k}_{21}' = \begin{bmatrix} -hv \operatorname{cosec} v & 0 & 0 \\ 0 & -\frac{Ttk}{L^2} & \frac{Qqk}{L} \\ 0 & -\frac{Qqk}{L} & Sck \end{bmatrix}, \quad (28)$$

with

$$\begin{aligned} k &= EI/L, \\ \eta &= Z/H, \\ S &= [(\alpha + \eta\beta)/\gamma] (\sin \beta \cosh \alpha - \eta \cos \beta \sinh \alpha), \\ Sc &= [(\alpha + \eta\beta)/\gamma] (\eta \sinh \alpha - \sin \beta), \\ Q &= (ZL/\gamma) [(\alpha - \eta\beta) (\cosh \alpha \cos \beta - 1) + (\beta + \eta\alpha) \sin \beta \sinh \alpha], \\ Qq &= (\eta/\gamma) (1 - s^2 p^2) (\alpha^2 + \beta^2) (\cosh \alpha - \cos \beta), \\ T &= (b^2/\alpha\beta\gamma) (\alpha + \eta\beta) (\sinh \alpha \cos \beta + \eta \sin \beta \cosh \alpha), \\ Tt &= (b^2/\alpha\beta\gamma) (\alpha + \eta\beta) (\sinh \alpha + \eta \sin \beta), \\ \gamma &= 2\eta(1 - \cosh \alpha \cos \beta) + (1 - \eta^2) \sin \beta \sinh \alpha. \end{aligned} \quad (29)$$

The dynamic member stiffness matrices of all the individual members may now be assembled to form $\mathbf{K}$, in the usual manner.

J_0 can be found from

$$J_0 = \Sigma J_m, \quad (30)$$

where J_m is the number of natural frequencies of a component member, with its ends clamped, which have been exceeded by ω^* , and the summation extends over all the component members. From equations (26), a natural frequency of a member with its ends clamped occurs when one or more of the elements of the matrices of equations (28) become infinite. From equations (29), this will occur when either

$$\sin v = 0, \quad (v \neq 0) \quad (31)$$

or

$$2\eta(1 - \cosh \alpha \cos \beta) + (1 - \eta^2) \sin \beta \sinh \alpha = 0. \quad (32)$$

Equation (31) is satisfied when

$$v = i\pi, \quad (i = 1, 2, 3, \dots) \quad (33)$$

and hence J_a , the component of J_m arising from equation (31), is given by

$$J_a \text{ is the highest integer } < \frac{v}{\pi}. \quad (34)$$

Clearly it is not so simple to find the component of J_m arising from equation (32). However, as these roots are the natural frequencies of the beam which involve bending they can be obtained as follows.

Consider a simply supported beam with moments applied at its ends. From equations (26)–(28) its member equations are

$$\mathbf{p} = k \mathbf{B} \mathbf{d}, \quad (35)$$

where

$$\mathbf{p} = \begin{bmatrix} m_1 \\ m_2 \end{bmatrix}, \quad \mathbf{d} = \begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} S & Sc \\ Sc & S \end{bmatrix}. \quad (36)$$

A natural frequency of this beam occurs when $|\mathbf{B}|$ is zero: i.e., when

$$S^2(1 - c^2) = 0. \quad (37)$$

Substituting for S and c from equations (29) gives the following condition for a natural frequency:

$$\frac{(\alpha + \eta\beta)^2}{\gamma} \sin \beta \sinh \alpha = 0. \quad (38)$$

It may be shown that this equation is satisfied only when

$$\sin \beta = 0, \quad (\beta \neq 0), \quad (39)$$

which requires that

$$\beta = i\pi, \quad (i = 1, 2, 3 \dots). \quad (40)$$

Hence, as it can be shown that β increases as ω increases, the number of natural frequencies of the simply supported beam exceeded by ω^* is J_c where

$$J_c \text{ is the highest integer } < \frac{\beta}{\pi}. \quad (41)$$

With this simply supported beam treated as a complete structure, for which the stiffness matrix is $\mathbf{B}$, equation (2) then gives

$$J_b = J_c - s\{\mathbf{B}\}, \quad (42)$$

where J_b is the component of J_m corresponding to flexural modes.

It can readily be shown that

$$s\{\mathbf{B}\} = \frac{1}{2}\{2 - \operatorname{sgn}[S(1 - c^2)] - \operatorname{sgn}[S]\}, \quad (43)$$

where $\operatorname{sgn}[S(1 - c^2)]$ and $\operatorname{sgn}[S]$ are $+1$ or -1 depending on the signs of $S(1 - c^2)$ and S , respectively.

Hence

$$J_m = J_a + J_b \quad (44)$$

and J_0 follows from equation (30).

J may now be calculated from equation (2). Thus a computer convergence routine can be written which adjusts ω^* systematically to converge with certainty upon any specified natural frequency of the frame.

Finally, the effects of axial load, rotatory inertia and shear deflection are uniquely described by the non-dimensional parameters p , r and s , respectively, and any particular effect may be neglected by setting the relevant parameter equal to zero.

3. EXPERIMENTAL RESULTS

Only the lower natural frequencies are usually of interest for civil engineering structures, and members are generally slender enough for their rotatory inertias and shear deflections to have little effect at such frequencies. The experimental results were obtained in this context, and thus show the effects of axial forces in members on the natural frequencies of the structure, and exclude significant effects of rotatory inertia and shear deflection.

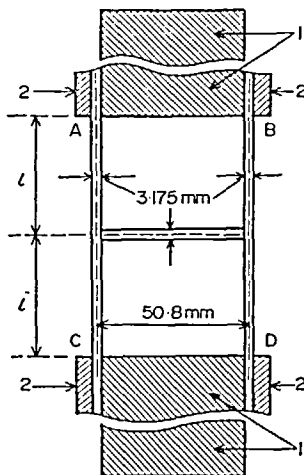


Figure 3. Side view of frame showing, in a simplified manner, the solid steel end mounting blocks (1) and the 12.7 mm thick steel plates (2) used to clamp the vertical members of the frame against the end blocks.

The model tested was the simple H-shaped frame shown in Figure 3, which is effectively built-in at A, B, C and D. The frame was fabricated from a single length of 25.4×3.175 mm ground steel stock which was cut into two equal lengths and a shorter length. The shorter length was soft soldered to the centres of the longer lengths to make the frame shown. The relatively large width of the members, 25.4 mm, ensured out-of-plane stability when the frame was loaded. The clamping arrangement enabled the end mounting blocks to be moved to vary l , so that the same model could be used to give results for what are, essentially, several different structures (see Figure 4). The frame was loaded by placing the frame and attached end mounting blocks centrally in a laboratory compression testing machine which had a load holding facility. Thus the vertical members carried equal axial compressive forces and the horizontal member was not axially loaded.

The unusual shape of the frame was suggested by the requirement that some members should carry high compressive loads which had to be applied without sway of the frame being affected by restraints or inertias due to the loading system.

A small piezoelectric gauge was used to apply an excitation to the frame which varied sinusoidally with time. As the excitation frequency was varied the natural frequencies were found by observing the peaks of response obtained from four small piezoelectric sensor gauges. These gauges were stuck onto the frame at positions which were chosen so that examination of the relative phase of their responses enabled those modes which involved in-plane flexure of the members to be separated from other modes and identified as being either symmetric or anti-symmetric about the vertical axis of symmetry of Figure 3.

The frame was tested with three specific values of l being used, corresponding to column

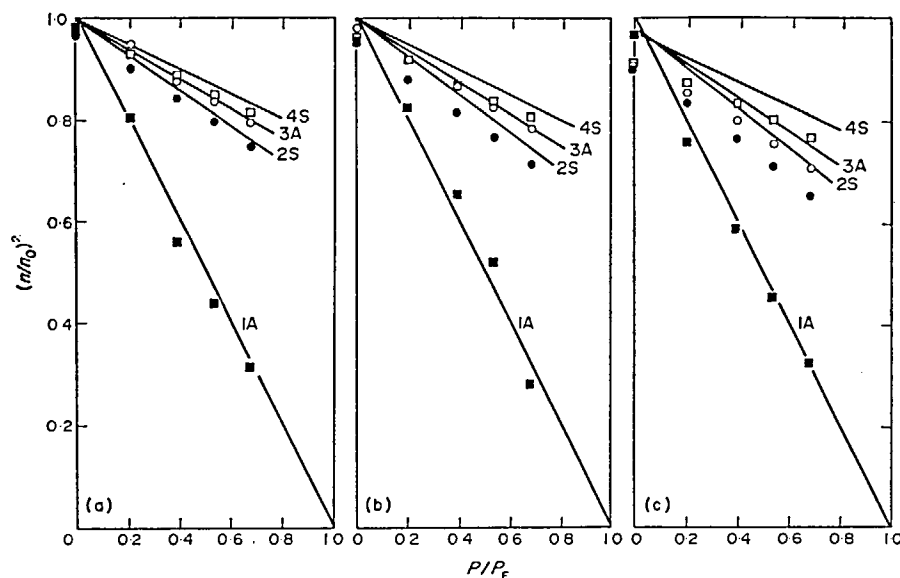


Figure 4. $(n/n_0)^2$ versus (P/P_E) . ■, ●, ○ and □ are experimental values for n , the i th natural frequency, for $i = 1, 2, 3$, and 4 respectively. Line i shows corresponding results computed by using correct values for p , r and s . For each i value all n values are divided by n_0 , the value of the i th natural frequency computed with $p = r = s = 0$. Experimental and calculated modes were antisymmetric (A) about the vertical axis of symmetry for i odd, and symmetric (S) for i even. (a) Column slenderness ratio of 200. (At $P/P_E = 0$, the experimental values of $(n/n_0)^2$ are 0.974 for □, 0.969 for ○, 0.968 for ● and 0.981 for ■.) (b) Column slenderness ratio of 140. (At $P/P_E = 0$, the experimental values of $(n/n_0)^2$ are 0.957 for □, 0.973 for ○, 0.947 for ● and 0.952 for ■.) (c) Column slenderness ratio of 80. (At $P/P_E = 0$, the experimental values of $(n/n_0)^2$ are 0.912 for □, 0.906 for ○, 0.902 for ● and 0.968 for ■.)

slenderness ratios† of 200, 140 and 80. For each slenderness ratio, loads causing axial forces, P , in the columns, of $0, 0.192 P_E, 0.385 P_E, 0.529 P_E$ and $0.673 P_E$ were applied in turn ($P_E = \pi^2 EI/l^2$). The first four in-plane, flexural natural frequencies of the frame were obtained in each case. The corresponding theoretical natural frequencies were calculated, first with the effects of axial load, rotatory inertia and shear deflection included (i.e., p, r and s with their correct values) and then with the three effects omitted (i.e., $p = r = s = 0$). The properties of the steel were $E = 212.95 \text{ GN/m}^2$, $G = 85.88 \text{ GN/m}^2$ and $\rho = 7829 \text{ kg/m}^3$. The value of k' for a rectangular section was taken as $2/3$ [15].

Figure 4 shows the experimental and theoretical variation of $(n/n_0)^2$ with P/P_E when the columns had slenderness ratios of 200, 140 and 80. The results clearly show the effect of axial load on the lower natural frequencies of the frame. The experimental results are in fairly good agreement with theory and become less accurate as the column slenderness ratio decreases, presumably due mainly to the increasing effect of the imperfect clamping. The computed results show that the theoretical lines form curves which are so shallow that they appear as straight lines on Figure 4.

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† The column slenderness ratio is defined as $l/\sqrt{I/A}$ where l is defined in Figure 3, and I is the relevant second moment of area of a member.

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APPENDIX I

In the case $b^2 r^2 s^2 > 1$, which was ignored in section 2, equation (18) yields an imaginary value for α , as follows:

$$\frac{\alpha'}{\beta} = \frac{b}{[2(1-s^2 p^2)]^{\frac{1}{2}}} \left\{ \Delta \mp \left[\Delta^2 + \frac{4}{b^2} (1-s^2 p^2) (1-b^2 r^2 s^2) \right]^{\frac{1}{2}} \right\}^{\frac{1}{2}}, \quad (\text{A1})$$

where

$$\alpha' = j\alpha \quad (\text{A2})$$

and

$$j = \sqrt{-1}. \quad (\text{A3})$$

Hence, substituting $\alpha = -j\alpha'$, equations (28) become

$$\mathbf{k}_{11} = \begin{bmatrix} hv \cot v & 0 & 0 \\ 0 & \frac{T'k}{L^2} & \frac{Q'k}{L} \\ 0 & \frac{Q'k}{L} & S'k \end{bmatrix},$$

$$\mathbf{k}_{22} = \begin{bmatrix} hv \cot v & 0 & 0 \\ 0 & \frac{T'k}{L^2} & -\frac{Q'k}{L} \\ 0 & -\frac{Q'k}{L} & S'k \end{bmatrix}, \quad \mathbf{k}_{12} = \mathbf{k}_{21}^t = \begin{bmatrix} -hv \operatorname{cosec} v & 0 & 0 \\ 0 & -\frac{T'k}{L^2} & \frac{Q'k}{L} \\ 0 & -\frac{Q'k}{L} & S'c'k \end{bmatrix}, \quad (\text{A4})$$

where

$$\begin{aligned}
 \eta' &= Z/H', \\
 H' &= [-\alpha'^2(1-s^2p^2) + b^2s^2]/\alpha'L, \\
 S' &= [(\alpha' + \eta'\beta)/\gamma'] (\sin \beta \cos \alpha' + \eta' \cos \beta \sin \alpha'), \\
 S'c' &= -[(\alpha' + \eta'\beta)/\gamma'] (\eta' \sin \alpha' + \sin \beta), \\
 Q' &= (ZL/\gamma') [(\alpha' - \eta'\beta) (\cos \alpha' \cos \beta - 1) + (\beta - \eta'\alpha') \sin \beta \sin \alpha'], \\
 Q'q' &= -(\eta'/\gamma') (1-s^2p^2) (\alpha'^2 - \beta^2) (\cos \alpha' - \cos \beta), \\
 T' &= (b^2/\alpha'\beta\gamma') (\alpha' + \eta'\beta) (\sin \alpha' \cos \beta + \eta' \sin \beta \cos \alpha'), \\
 T't' &= (b^2/\alpha'\beta\gamma') (\alpha' + \eta'\beta) (\sin \alpha' + \eta' \sin \beta), \\
 \gamma' &= 2\eta'(1 - \cos \alpha' \cos \beta) + (1 + \eta'^2) \sin \alpha' \sin \beta.
 \end{aligned} \tag{A5}$$

The manner in which J_b is obtained remains unchanged, but now the stiffness matrix of equation (36) is given by

$$\mathbf{B} = \begin{bmatrix} S' & S'c' \\ S'c' & S' \end{bmatrix} \tag{A6}$$

and hence a natural frequency of the simply supported beam occurs when

$$S'^2(1 - c'^2) = 0. \tag{A7}$$

Substituting for S' and c' from equations (A5) gives the following condition for a natural frequency:

$$\frac{(\alpha' + \eta'\beta)^2}{\gamma'} \sin \beta \sin \alpha' = 0. \tag{A8}$$

This equation is only satisfied when either

$$\sin \beta = 0, \quad (\beta \neq 0) \tag{A9}$$

or

$$\sin \alpha' = 0, \quad (\alpha' \neq 0). \tag{A10}$$

Hence equations (A9) and (A10) will be satisfied when

$$\beta = i\pi, \quad (i = 1, 2, 3 \dots) \tag{A11}$$

and

$$\alpha' = i\pi, \quad (i = 1, 2, 3 \dots), \tag{A12}$$

respectively. Thus, as it can be shown that β and α' increase as ω increases, J_c is now given by

$$J_c = J_d + J_e, \tag{A13}$$

where

$$J_d \text{ is the highest integer } < \frac{\beta}{\pi},$$

$$J_e \text{ is the highest integer } < \frac{\alpha'}{\pi}.$$

$$\tag{A14}$$

The value of J_b may now be obtained by substituting equation (A13) into equation (42), where $s\{\mathbf{B}\}$ is given by

$$s\{\mathbf{B}\} = \frac{1}{2}\{2 - \operatorname{sgn}[S'(1 - c'^2)] - \operatorname{sgn}[S']\}. \quad (\text{A15})$$

The values of J_a , J_m and J_0 can then be calculated as described in section 2.

APPENDIX II

NOTATION

Except where the context dictates otherwise, the words "displacement", "force", "moment", etc., are understood to mean "the amplitude of the sinusoidally varying displacement", etc. Likewise the terms stiffness, stiffness matrix, etc., are understood to mean "dynamic stiffness" etc.

A	cross-section area of member
b	$\omega L^2 \sqrt{\rho A/EI}$
c	stiffness function defined by equation (29)
C_1, C_2, C_3, C_4	constants of integration
C_1', C_2', C_3', C_4'	constants of integration related to C_1 , etc., by equation (21)
d_1, d_2	$\{\delta_{x1}, \delta_{y1}, \psi_1\}$ and $\{\delta_{x2}, \delta_{y2}, \psi_2\}$, respectively
$\mathbf{D}$	overall nodal displacement vector "corresponding" to $\mathbf{K}$, in frame coordinates
E	Young's modulus
G	modulus of rigidity
h	EA/L
H	$[(1 - s^2 p^2)x^2 + b^2 s^2]/\alpha L$
I	second moment of area of a member cross-section about the axis of rotation
J	number of natural frequencies of the structure exceeded by ω^*
J_0	number of natural frequencies of the structure with $\mathbf{D} = \mathbf{0}$ exceeded by ω^*
J_m	$J_m = J_a + J_b$
J_a	number of longitudinal natural frequencies of a member with its ends clamped exceeded by ω^*
J_b	number of in-plane flexural natural frequencies of a member with its ends clamped exceeded by ω^*
k	EI/L
k'	section shape factor
$\mathbf{k}_{11}, \mathbf{k}_{12}$ $\mathbf{k}_{21}, \mathbf{k}_{22}$	member stiffness matrices, referred to member axes
$\mathbf{K}$	overall stiffness matrix
$ \mathbf{K} $	determinant of the matrix $\mathbf{K}$
l	length of a column (see Figure 3)
$l/\sqrt{I/A}$	column slenderness ratio (see Figure 3)
L	length of a member (see Figure 2)
m_1, m_2	moments at ends 1 and 2 of a member, respectively (see Figure 2)
n	natural frequency when p, r and s assume their correct values
n_0	natural frequency when $p = r = s = 0$
p	$\sqrt{PL^2/EI}$
$\mathbf{p}_1, \mathbf{p}_2$	$\{p_{x1}, p_{y1}, m_1\}$ and $\{p_{x2}, p_{y2}, m_2\}$, respectively
p_{x1}, p_{x2}	axial forces at ends 1 and 2 of a member, respectively (see Figure 2)
p_{y1}, p_{y2}	shear forces at ends 1 and 2 of a member, respectively (see Figure 2)
P	static compressive axial load
P_E	Euler load of a pin ended column ($= \pi^2 EI/l^2$)
q	stiffness functions defined by equation (29)
r	$\sqrt{I/AL^2}$
s	$\sqrt{EI/\phi L^2}$

$s\{ \}$	the sign count of the matrix within the bracket (see below equation (2))
$\text{sgn}[\]$	this symbol takes the value +1 if the expression within the bracket is positive and -1 if the expression is negative
$\left. \begin{matrix} S \\ t \\ T \end{matrix} \right\}$	stiffness functions defined by equation (29)
x, y	member axes (see Figure 2)
Z	$[(1 - s^2 p^2) \beta^2 - b^2 s^2] / \beta L$
α	defined by equation (18)
β	defined by equation (29)
γ	defined by equation (29)
$\left. \begin{matrix} \delta_{x1}, \delta_{x2} \\ \delta_{y1}, \delta_{y2} \end{matrix} \right\}$	translations at the ends of a member (see Figure 2)
Δ	$(p^2/b^2) + r^2(1 - s^2 p^2) + s^2$
η	Z/H
v	$\omega L \sqrt{\rho/E}$
ξ	x/L
ρ	density of the member material
ϕ	$k'AG$
Ψ_1, Ψ_2	rotations due to bending at ends 1 and 2 of a member, respectively (see Figure 2)
ω	(circular) frequency
ω^*	a trial value of ω